

BUSINESS ANALYSIS & REQUIREMENTS SPECIFICATION

PK Fokam Research Center | ENSPY 4GI

MICROFI

Digital Platform for Itinerant Collection
and Management of Microfinance Institutions

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FRAMEWORK & VERSION

Research & Innovation Hub

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1. Strategic Framework & Business Environment

Institutional Context (PK Fokam Research Center & MFIs)

The **MICROFI** project is developed under the aegis of the PK Fokam Research Center in partnership with Microfinance Institutions (MFIs) across the CEMAC zone. The strategic objective is to design a standardized, modular, and agnostic technology infrastructure to address the challenges of rural and semi-urban financial inclusion.

The CEMAC zone is characterized by a strong duality in the microfinance sector:

CATEGORY 1 — Cooperatives & Mutual Networks

MC² / MUFFA networks, village cooperatives, and GICs, historically operating on decentralized foundations with a heavy reliance on paper and infrastructure constraints (intermittent connectivity, rural remoteness).

CATEGORY 2 — Corporations & Centralized MFIs

Regulated commercial MFIs equipped with centralized Information Systems (Sopra Amplitude, Kovara, Asengo) but suffering from rigidity in expanding their proximity collection channels.

Stakeholder Mapping (Stakeholders Analysis & RACI Matrix)

The success of the project depends on rigorous coordination among the various internal and external actors of the financial ecosystem.

RACI legend: R = Responsible (executes the task), A = Accountable (validates / accountable), C = Consulted (provides expertise), I = Informed (kept up to date).

Business Risk Analysis & Regulatory Constraints (COBAC / CEMAC)

Proximity collection activity in the CEMAC zone is governed by strict requirements set out by COBAC and BEAC regarding anti-money laundering, fund security, and customer protection.

CASH TRACEABILITY

Obligation to link every physical collection to an identified client and to ensure rigorous traceability of banknote denominations (BEAC).

CASH-HANDLING EXPOSURE CEILINGS

High exposure of field agents to physical assault and cash theft.

COBAC COMPLIANCE

Requirement for daily accounting reconciliation and a formal ban on holding unrecorded funds at the end of the day.

Stakeholder	Role & main responsibility	RACI
Collection Agent (TPU)	Field operator covering markets and counters. Records deposits, enters denomination breakdowns, prints thermal receipts.	A/R
Branch Cashier	Sedentary agent managing the branch's physical cash. Opens sessions, performs end-of-day reconciliation (OFJ), validates deposits.	A/R
Branch Manager / Director	Responsible for operational performance and security. Geolocated tracking of routes, approval of escrow ceilings, granting of grace extensions (<i>Bouton Prolonger</i>).	A/C
System Administrator	Central technical team (MFI IT). Manages access rights, enrolls terminals, configures schedule profiles.	R/C/A
Regulator (COBAC / BEAC)	Monetary and banking supervisory authority. Regulatory compliance, adherence to prudential standards, and flow traceability.	I
Client (Member / Merchant)	End-user of the proximity collection service. Activates and renews digital profile (tokenized booklet), consults personal profile, account balance and contribution history via the mobile application or USSD.	I/R

Table 1: Stakeholders and RACI Matrix

2. Analysis of the Existing Process (AS-IS)

Mapping of the Manual Field Collection Process

The current operating model (AS-IS) relies almost exclusively on paper media and cumbersome physical processes, broken down into three chronological phases:

★ MORNING	★ FIELD	★ EVENING
The agent reports to the branch to collect their counterfoil receipt book, receive manual instructions, and check their route history on paper.	The agent covers market stalls, collects cash from merchants, manually writes a carbon-copy receipt, tears it off, and stamps the client's passbook.	The agent returns to the branch with their bag of cash. The cashier manually counts all the cash and cross-checks it against the passbooks (duration: 1.5 to 2.5 hours of congestion).

Modeling of the “As-Is” Process (Actors & Flows)

The text diagram below formalizes the interactions and friction points of the legacy model:

Client (Merchant) → Hands over cash and presents their paper passbook → **Field Agent**
→ Fills out and signs a carbon-copy receipt.

Field Agent (Evening) → Returns to the branch with the cash bags → **Branch Cashier**
→ Tedious manual counting + cross-checking of passbooks.

Outcome: Delays, disputes over counterfeit notes, unexplained variances, and accounting entries deferred to the next day.

Diagnosis of Dysfunctions & Bottlenecks

An audit of current practices reveals five major systemic failures:

- **Opacity of collection & “cavalerie” risk** — In the absence of immediate digital traceability, some agents use funds collected from Client B to cover a late payment or unauthorized disbursement owed by Client A (a form of check-kiting / rolling shortfall).
- **Slow Evening reconciliation / cash counting** — Manual counting of heterogeneous masses of banknotes by cashiers creates enormous queues and human summation errors.
- **Counterfeit notes and disputes** — When a counterfeit note is detected in the evening at the branch, it is impossible to attribute it to a specific client, creating financial disputes and tension directed at the agent.
- **Lack of GPS visibility and escrow overruns** — Agents often collect beyond their actual financial guarantee without automated control, exposing the MFI to outright losses in the event of misappropriation or theft.
- **Information asymmetry** — Client account balances on the Core Banking System (CBS) remain stale throughout the entire day of field operations.

Competitive Technology Landscape

To strategically position MICROFI within the microfinance landscape of Cameroon and the CEMAC zone, this section presents a comparative analysis of the technologies currently used in the field, followed by **MICROFI**'s unique positioning.

The proximity-collection market today relies on three major technology families.

A — Dedicated Payment Terminals (POS / Portable POS)

Standalone physical devices distributed to agents (e.g., traditional pre-smartphone solutions).

Advantages: Very robust physically; natively integrate a thermal printer to deliver an immediate paper receipt (a major trust factor for older or illiterate merchants).

Disadvantages: Very high acquisition and hardware-maintenance cost for the MFI; limited functionality; absolute dependence on a data SIM card; no advanced GPS tracking.

B — Local Fintech Smartphone Apps (e.g., Smobilpay, Blucash)

Android mobile applications connected to Bluetooth mini-printers.

Advantages: Flexible and field-adapted (offline mode often available); reduced hardware cost (the agent can use their own smartphone).

Disadvantages: Often disconnected from the MFI’s accounting system (requires catch-up data entry); escrow management still very manual or limited to a single banking partner.

C — Direct USSD Channels from Telecom Operators (Push & Pull / Mobile Money)

The client saves on their own by dialing a code (e.g., *126# or *150#) from their own phone.

Advantages: Zero risk of physical cash theft; no field-agent logistics required; 24/7 availability.

Disadvantages: High transaction fees charged by operators (MTN/Orange) that discourage small daily savings; dehumanization of the relationship (market merchants prefer the daily visit of a trusted human collector).

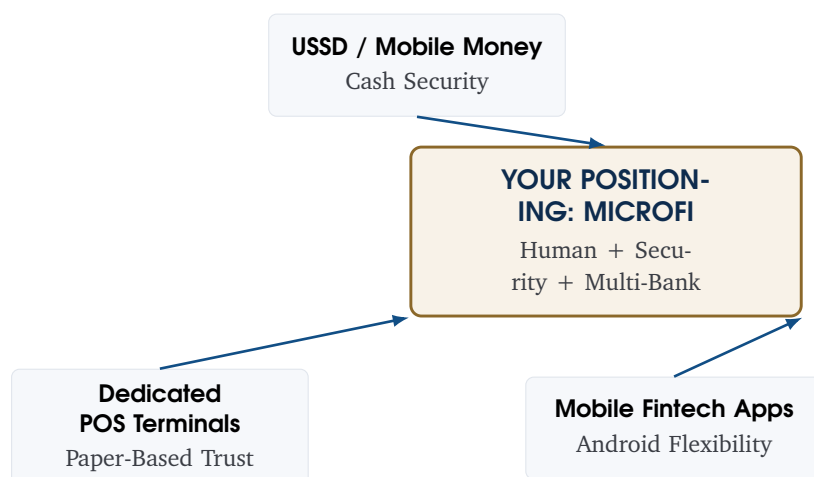
Summary Comparison

Technology		Cost for MFI	Cash Security	Customer Experience	System Integration
Dedicated Terminal	POS	● High	● Medium (cash transported)	● Excellent (paper)	● Limited
Mobile Apps	Fintech	● Low	● Medium (cash transported)	● Good (SMS + Bluetooth)	● Partial
USSD / Mobile Money		● Medium (fees)	● Absolute (0 cash)	● Weak (complex/isolated)	● Good (API)

Table 2: Comparative Summary of Current Collection Technologies

MICROFI’s Unique Positioning (Value Proposition)

The PK Fokam Research Center’s solution positions itself as a **third-generation hybrid platform**. It does not choose between the human touch and digital efficiency — it merges the best of all three worlds.



Your competitive advantage rests on three pillars of disruption:

PILLAR 1

“Zero-Cash” for the Institution via the Universal Escrow Wallet — Unlike classic fintechs that merely digitize the receipt, MICROFI uses the GIMAC switch and partner banks to lock guarantees in a fully digital way. The MFI is covered against fraud before the agent even begins their route.

PILLAR 2

Total Agnosticism (Multi-Tenant & Multi-Bank) — The solution is not locked into a single operator or bank. It offers a standardized technical layer sellable to any MFI (Category 1 or 2), guaranteeing the institution’s full sovereignty over its data.

PILLAR 3

Absolute Operational Control (Strict Tracking & Denomination Breakdown) — By natively integrating intelligent GPS tracking (NFR-09) and a mandatory full denomination breakdown from 500 XAF upward (FR-08), MICROFI gives risk managers a level of control never achieved by competing solutions.

3. Definition of the Target Process (TO-BE)

Vision of the Digitized Process

The target process (TO-BE) introduces the **MICROFI** platform, built around secure mobile terminals (TPU) and a centralized Back-Office portal. The objective is to eliminate paper, automate cash controls, and interface field operations in real time with the MFI’s central system.

Modeling of the “To-Be” Process (Digital Workflow)

The collection cycle is fully secured and streamlined:

★ MORNING

Biometric or PIN login by the agent on the TPU. Automatic verification of the digital escrow (*Digital Escrow Engine*). If the escrow ceiling has been reached, the terminal blocks any new collection until a deposit has been made.

★ FIELD

Client QR-code scan or instant lookup. Entry of the amount and mandatory denomination breakdown (BEAC breakdown: 10k, 5k, 2k, 1k, 500F, and coins). Immediate printing of the thermal receipt via Bluetooth.

★ EVENING — DIGITAL CASH DESK / OFJ

Data synchronization. The cashier enters the physical amounts counted by denomination; the system instantly calculates the variance ($\Delta = \text{Physical} - \text{Digital}$). One-click validation and automatic posting into the CBS.

Impact Matrix of Changes per Actor

Actor	Impact on daily work (As-Is → To-Be)	Main gain
Collection Agent	Replacement of the paper passbook with a rugged Android terminal. Mandatory structured data entry.	Elimination of calculation errors, speed, and offline autonomy.
Branch Cashier	End of blind counting and manual passbook reconciliation. Batch digital validation.	Reduction of end-of-day closing time from several hours to some few minutes.
Branch Manager	Live tracking of routes via geolocation and oversight of outstanding escrow balances.	Complete control over fraud risk and mandate overruns.
Client (Member)	Paper passbook replaced by digital profile + QR card. Can consult balance and contribution history on smartphone (or USSD). Annual digital booklet renewal via agent.	Real-time visibility of savings, elimination of lost/damaged paper booklets, continuous proof of deposits.

Table 3: Change Impact Matrix

Definition of Business Rules

BR-01

Cash Variance Calculation / Digital Cash Desk — The variance Δ is calculated automatically by the system during the evening reconciliation. Any negative variance ($\Delta < 0$) blocks session closing and immediately generates an audit ticket for internal control.

BR-02

Denomination Breakdown Rules — Entry of the denomination breakdown (10,000, 5,000, 2,000, 1,000, 500 XAF, and coins) is mandatory for any deposit above a defined threshold. The sum of the breakdown must strictly match the total amount entered.

BR-04

Digital Booklet Tokenization & Agent-Sponsored Activation — Existing MFI members shall activate their mobile profile using a temporary Activation ID issued by the Core Banking System. Activation (and annual renewal) requires sponsorship by a registered Field Agent and payment of a configurable fee that is automatically split between the agent (commission) and the MFI. The resulting digital access token is valid for 365 days. Upon expiry the mobile profile enters a read-only state (balances remain visible; transactional operations are locked) until renewed. The system shall never create new customer records inside the Core Banking System.

BR-05

Mandatory GPS for Collection — No collection transaction shall be initiated or validated if the agent terminal does not have an active GPS/location service providing a current fix. If GPS is disabled, denied by the operating system, or unable to produce a valid position, the mobile application shall block the collection flow and display a clear message requiring the agent to enable location services. Network-based approximate location alone is not sufficient to authorize a new collection.

4. Business & Technical Requirements Specification

Detailed Functional Requirements (FR)

The matrix below details the complete set of functional requirements identified for the MICROFI platform, organized by functional module. **All requirements are listed here in full**; MVP scoping and release prioritization will be addressed separately in a subsequent, dedicated feature-priority matrix.

MODULE 1 — Profile Management & Access Security

FR-01

Single Sign-On Authentication — The system shall authenticate the field agent via a unique identifier tied to the hardware address (IMEI) of their smartphone.

FR-02

Admin Enrollment — The MFI administrator shall be able to create, modify, suspend, or delete an agent's profile from the central console.

MODULE 2 — Escrow & Financial Lock Management

FR-03

Multi-Bank Top-Up — The system shall allow the agent's escrow Wallet to be topped up via Mobile Money (Orange/MTN) or interbank transfer (GIMAC).

FR-04

Dynamic Lockout — The system shall automatically block entry of new transactions as soon as the cumulative cash collected reaches the active escrow amount.

FR-05

Exceptional Adjustment — The administrator shall be able to temporarily modify an agent's collection ceiling remotely during the course of the day.

MODULE 3 — Collection Operations & Proof of Deposit

FR-06

Multi-Method Client Identification — The agent shall be able to identify a client through all supported identification channels defined by the business process: QR-Code scanning from the membership card/passbook, manual entry of the client identifier, search by registered phone number, and assisted search using client information available in the customer registry (name or reference data). The system shall return a unique client profile before any collection transaction is initiated.

FR-07

Offline Mode — The mobile application shall allow secure local recording of collections in areas without network coverage.

FR-08

Full Denomination Breakdown — The system shall require the agent to enter the physical breakdown of all BEAC denominations without exception (500, 1,000, 2,000, 5,000, 10,000 XAF), plus a single aggregate entry line for coins.

FR-09

Multi-Channel Notification — Upon validation, the system shall trigger a Flash SMS to the client and allow printing of a receipt via a Bluetooth printer.

MODULE 4 — Field Tracking & Security**FR-10**

Real-Time Geolocation — The system shall capture and send the agent's GPS coordinates to the MFI Back-Office at regular intervals (e.g., every 5 minutes) while the application is active.

FR-11

Route Mapping — The administrator Back-Office shall display each agent's historical daily route on a dynamic map to verify actual coverage of assigned markets.

FR-12

Transaction Timestamping & Mandatory Geotagging — For each deposit collected, the system shall invisibly record the precise GPS coordinates and UTC timestamp of the transaction location. **Collection is forbidden unless GPS/location services are active and a valid fix is available** (see BR-05). The geotag is non-editable by the agent.

FR-13

Active Geofencing Alert — The system shall generate an immediate visual alert on the administrator console if an agent conducts a transaction or moves outside their assigned perimeter (market/district).

FR-14

Distress / SOS Button — The mobile application shall include a discreet emergency button allowing the agent, in case of assault, to instantly send their exact GPS position via SMS and Internet to their branch administrator.

MODULE 5 — End-of-Day Closing (OFJ) & Reporting**FR-15**

Dynamic Schedule Management — The administrator shall be able to configure session opening and closing time windows tailored to each branch.

FR-16

End-of-Day Reconciliation — The system shall provide a reconciliation interface between the physical cash declared via denomination breakdown and the application's accounting total.

FR-17

Variance Regularization — The system shall allow the administrator to record a cash variance as an “agent debt” on their profile.

FR-18

Core Banking Export — The system shall generate a daily entries file compatible with major Core Banking software (Amplitude, FinanSoft).

MODULE 6 — Client Digital Activation, Profile & Self-Service (Digital Booklet)**FR-19**

Tokenized Digital Activation & Annual Renewal (Digital Booklet) — The system shall enable existing MFI members to activate (and annually renew) their mobile profile using a temporary Activation ID issued by the Core Banking System. Activation requires sponsorship by a registered Field Agent and payment of a configurable fee that is automatically split between the agent (commission) and the MFI. Upon successful payment the system issues a digital access token valid for 365 days. When the token expires the profile enters a read-only state (balances remain visible; transactional operations from the mobile application are locked) until a new agent-sponsored renewal occurs. The system shall never create new customer records inside the Core Banking System; non-members are redirected to physical branch/agent registration.

FR-20

Client Profile Consultation — Once activated, the client shall be able to view and consult his/her personal profile on the mobile application: full name, membership number, photo (if available), registered phone number, assigned branch, activation/renewal status and token

expiry date.

FR-21

Account Balance & Summary — The client shall be able to consult in real time (or last synchronized) the current balance of his/her savings account(s), the cumulative amount collected year-to-date, and a high-level account summary. Balance visibility remains available even when the digital token is expired (read-only mode).

FR-22

Contribution History & Mini-Statement — The client shall be able to consult the chronological history of his/her contributions (date, amount, agent reference, location if available) with filtering by period. A short mini-statement of the last N transactions shall also be accessible via USSD or Flash SMS for clients using basic feature phones.

User Stories & Use Cases

For each functional requirement, a corresponding user story (in standard Agile format) and a structured use-case description are provided below. Use cases follow a consistent template: **ID, Name, Actors, Objectives, Preconditions, Postconditions, Nominal Scenario, Alternative Scenarios, and Business Rules.**

MODULE 1 — Profile Management & Access Security

FR-01 — Single Sign-On Authentication

User Story (US-01): As a Collection Agent, I want to authenticate on the TPU using a unique identifier bound to the device IMEI so that only authorized terminals can access collection functions and unauthorized devices are blocked.

UC-01

Name: Agent Authentication via IMEI-bound SSO

Actors: Collection Agent (primary), System

Objectives: Authenticate the agent securely and bind the session to the enrolled hardware terminal.

Preconditions: The agent's profile has been enrolled by an Administrator; the device IMEI is registered; the agent has a valid PIN or biometric credential.

Postconditions: A secure session is opened; the agent can access authorized modules; an audit log entry is created.

Nominal Scenario:

1. Agent launches the MICROFI mobile application.
2. System reads the device IMEI and prompts for PIN/biometric.
3. Agent enters credentials.

4. System validates IMEI against the enrolled terminal and credentials against the agent profile.
5. System opens a session and displays the main collection dashboard.

Alternative Scenarios:

- *A1 — Unknown IMEI:* System rejects login and displays “Device not enrolled”. Agent contacts Administrator.
- *A2 — Invalid credentials:* After 3 failed attempts the account is temporarily locked; Administrator is notified.
- *A3 — Offline authentication:* System validates against locally cached credentials (encrypted) and marks the session for later online verification.

Business Rules: BR-Auth-01: Session lifetime is limited to the configured working window (see FR-15). BR-Auth-02: IMEI binding is mandatory; no soft-login without hardware association.

FR-02 — Admin Enrollment

User Story (US-02): As a System Administrator, I want to create, modify, suspend or delete agent profiles from the central console so that only authorized personnel can operate collection terminals.

UC-02

Name: Agent Profile Enrollment & Lifecycle Management

Actors: System Administrator (primary), Branch Manager (secondary)

Objectives: Maintain an accurate and secure registry of collection agents and their assigned terminals.

Preconditions: Administrator is authenticated on the Back-Office; the target branch exists.

Postconditions: Agent profile is created/updated/suspended; associated terminal (IMEI) is linked or unlinked; change is audited.

Nominal Scenario:

1. Administrator navigates to Agent Management.
2. Administrator fills agent identity, role, branch assignment, escrow guarantee reference and device IMEI.
3. System validates uniqueness of identifier and IMEI.
4. Administrator confirms; system creates the profile and activates the terminal.
5. Confirmation notification is sent to the agent (optional SMS).

Alternative Scenarios:

- *A1 — Duplicate IMEI:* System rejects and displays existing assignment; Administrator must first unbind the previous agent.
- *A2 — Suspension:* Administrator selects “Suspend”; system immediately invalidates active sessions on that terminal.
- *A3 — Deletion:* Soft-delete only; historical transactions remain linked for audit.

Business Rules: BR-Enroll-01: Only Administrators with “Agent Management” privilege can perform CRUD. BR-Enroll-02: An agent cannot be deleted while open sessions or unresolved variances exist.

MODULE 2 — Escrow & Financial Lock Management

FR-03 — Multi-Bank Top-Up

User Story (US-03): As a Collection Agent (or Administrator), I want to top up my digital escrow wallet via Mobile Money or interbank transfer so that I can increase my authorized collection ceiling without returning to the branch.

UC-03

Name: Escrow Wallet Top-Up (Multi-Channel)

Actors: Collection Agent, Administrator, Payment Gateway (GIMAC / Mobile Money)

Objectives: Increase the agent's available escrow balance securely and in near real-time.

Preconditions: Agent profile is active; at least one payment channel is configured for the MFI.

Postconditions: Escrow balance is increased; lockout threshold is updated; transaction is logged and visible on Back-Office.

Nominal Scenario:

1. Agent (or Admin) initiates top-up from the Wallet screen.
2. Agent selects channel (Orange Money, MTN MoMo, or GIMAC transfer) and enters amount.
3. System generates a payment reference and redirects or displays USSD/instructions.
4. Payment provider confirms the transfer.
5. System credits the escrow wallet and raises the collection ceiling.

Alternative Scenarios:

- *A1 — Payment failure:* System leaves balance unchanged and notifies the agent; retry is possible.
- *A2 — Offline initiation:* Request is queued; credit is applied only after successful online confirmation.
- *A3 — Manual cashier top-up (MVP fallback):* Cashier records a physical deposit; Administrator validates and credits the wallet.

Business Rules: BR-Escrow-01: Top-up amounts must be positive and within configured limits. BR-Escrow-02: Full multi-bank automation is backlog; MVP supports single-channel + manual recording.

FR-04 — Dynamic Escrow Lockout

User Story (US-04): As a Risk Manager / System, I want the terminal to automatically block new collections once the cumulative cash reaches the active escrow amount so that the MFI is never exposed beyond the agent's guarantee.

UC-04

Name: Automatic Collection Lockout on Escrow Exhaustion

Actors: System (primary), Collection Agent

Objectives: Prevent collection beyond the agent's financial guarantee in real time.

Preconditions: Agent is authenticated; an active escrow balance exists; a collection transaction is being initiated.

Postconditions: If cumulative cash $\geq$ escrow, further collections are blocked until a deposit or top-up occurs; agent is notified.

Nominal Scenario:

1. Agent attempts to start a new collection.
2. System calculates current cumulative collected amount for the day.
3. System compares against remaining escrow balance.
4. If remaining escrow > 0 , collection proceeds; otherwise system displays lockout message and disables the collection button.

Alternative Scenarios:

- *A1 — Partial remaining balance:* System allows collection only up to the residual escrow amount.
- *A2 — Concurrent top-up:* If a top-up is confirmed while locked, the lock is released automatically.

Business Rules: BR-03 (from main document) applies. BR-Lock-01: Lockout is enforced both online and offline using the last synchronized escrow value.

FR-05 — Exceptional Ceiling Adjustment

User Story (US-05): As a Branch Manager or Administrator, I want to temporarily raise an agent's collection ceiling remotely so that exceptional high-volume days can be handled without forcing the agent to interrupt the route.

UC-05

Name: Remote Temporary Escrow Ceiling Override

Actors: Branch Manager / Administrator (primary), System

Objectives: Grant a temporary, audited increase of the collection ceiling.

Preconditions: Manager has the privilege; agent is active; justification can be recorded.

Postconditions: Temporary ceiling is applied for the current day (or configured duration); original ceiling is restored automatically; change is fully audited.

Nominal Scenario:

1. Manager opens the agent's live status on the Back-Office.
2. Manager selects "Adjust Ceiling", enters new temporary amount and reason.
3. System validates privilege and records the override.
4. New ceiling is pushed to the agent's terminal (or applied on next sync).
5. Agent receives a notification that the ceiling has been raised.

Alternative Scenarios:

- *A1 — Insufficient privilege:* Request is rejected; only Risk Director or designated roles can approve large increases.

- *A2 — Expiry:* At end of configured window the original ceiling is restored and any excess is locked.

Business Rules: BR-Adj-01: Every override must carry a free-text justification. BR-Adj-02: Maximum temporary increase is limited by MFI policy (configurable).

MODULE 3 — Collection Operations & Proof of Deposit

FR-06 — Multi-Method Client Identification

User Story (US-06): As a Collection Agent, I want to identify a client by QR code, client ID, phone number or name search so that I can quickly and accurately select the correct account even when the client has no physical card.

UC-06

Name: Multi-Channel Client Lookup

Actors: Collection Agent (primary), System

Objectives: Retrieve a unique, verified client profile before any collection is recorded.

Preconditions: Agent is authenticated; client registry is available (online or cached).

Postconditions: A single client profile is selected and locked for the transaction; any ambiguity is resolved before amount entry.

Nominal Scenario:

1. Agent chooses identification method (QR scan, manual ID, phone, or name search).
2. System queries the registry (local cache first if offline).
3. System returns matching profile(s).
4. Agent confirms the unique client; system proceeds to amount entry.

Alternative Scenarios:

- *A1 — Multiple matches:* System displays a disambiguation list (name + phone + photo if available); agent must select one.
- *A2 — No match:* System allows creation of a provisional client record (if policy permits) or aborts the collection.
- *A3 — Offline incomplete registry:* System searches only the locally cached subset and warns that results may be incomplete.

Business Rules: BR-ID-01: A collection cannot be validated without a resolved unique client. BR-ID-02: All identification attempts are logged for audit.

FR-07 — Offline Collection Mode

User Story (US-07): As a Collection Agent, I want to record collections securely even when there is no network coverage so that market operations are never interrupted by connectivity issues.

UC-07

Name: Secure Offline Collection Recording

Actors: Collection Agent, System (local encrypted store)

Objectives: Persist collection data locally with integrity and confidentiality until connectivity is restored.

Preconditions: Agent is authenticated; local encrypted storage is initialized; sufficient device storage remains; **GPS/location services are enabled and a valid fix is available (BR-05).**

Postconditions: Transaction is stored encrypted on device; marked as “pending sync”; appears in the agent’s local history.

Nominal Scenario:

1. Agent performs a collection while offline.
2. System writes the full transaction (client, amount, denomination breakdown, GPS, timestamp) to the encrypted local store.
3. System updates local cumulative totals and remaining escrow.
4. Agent receives visual confirmation that the transaction is saved offline.

Alternative Scenarios:

- *A1 — Storage full:* System refuses new offline transactions and prompts the agent to move to a coverage area to sync.
- *A2 — GPS disabled or no fix:* System blocks the collection flow and requires the agent to enable location services (BR-05); no offline record is written.
- *A3 — Connectivity restored:* Background sync uploads pending transactions in chronological order; conflicts are flagged for Administrator review.

Business Rules: NFR-01 (AES-256) applies. BR-05 (mandatory GPS). BR-Off-01: Offline transactions inherit the last known escrow ceiling. BR-Off-02: Sync is automatic and non-blocking (NFR-08).

FR-08 — Full Denomination Breakdown

User Story (US-08): As a Collection Agent / Risk Officer, I want every deposit above the threshold to require a complete BEAC denomination breakdown so that counterfeit notes can be traced and cash variance is minimized.

UC-08

Name: Mandatory Denomination Breakdown Entry

Actors: Collection Agent (primary), System

Objectives: Capture the exact physical composition of the cash received.

Preconditions: Client is identified; amount to collect is known; amount exceeds the configured threshold.

Postconditions: Breakdown is recorded; sum of denominations equals the declared total; transaction can proceed to validation.

Nominal Scenario:

1. System presents the denomination grid (10 000, 5 000, 2 000, 1 000, 500 XAF + coins aggregate).
2. Agent enters quantity for each denomination.
3. System calculates the implied total in real time.

4. Agent confirms; system verifies equality with the declared amount and stores the breakdown.

Alternative Scenarios:

- *A1 — Sum mismatch:* System blocks validation and highlights the discrepancy until corrected.
- *A2 — Below threshold:* Breakdown is optional; agent may skip.

Business Rules: BR-02 (main document) applies. BR-Denom-01: Coins are entered as a single aggregate value. BR-Denom-02: Breakdown is mandatory for BEAC traceability.

FR-09 — Multi-Channel Notification

User Story (US-09): As a Client / Collection Agent, I want an immediate Flash SMS and an optional Bluetooth thermal receipt after each validated deposit so that both parties have instant proof of the transaction.

UC-09

Name: Post-Validation Multi-Channel Notification

Actors: System, SMS Gateway, Bluetooth Printer, Client, Collection Agent

Objectives: Deliver instantaneous, multi-channel confirmation of a successful collection.

Preconditions: Transaction has been validated; client phone number is known (for SMS); printer is paired (for paper).

Postconditions: SMS is queued/sent; receipt is printed (if requested); notification status is logged.

Nominal Scenario:

1. Transaction is validated.
2. System composes Flash SMS with amount, date, agent reference and new balance (if available).
3. System sends SMS via gateway and, if printer is available, formats and prints the thermal receipt.
4. Agent and client receive confirmation.

Alternative Scenarios:

- *A1 — SMS failure:* Transaction remains valid; SMS is retried later; agent is informed.
- *A2 — No printer:* Only SMS is sent; agent can later reprint from history.
- *A3 — Offline:* Notifications are queued and dispatched on next successful sync.

Business Rules: NFR-05 (15 s latency target) applies when online. BR-Notif-01: Receipt content must include mandatory legal mentions (MFI name, amount, date, agent ID).

MODULE 4 — Field Tracking & Security

FR-10 — Real-Time Geolocation

User Story (US-10): As a Branch Manager, I want the agent's GPS position to be sent at regular intervals while the application is active so that I can monitor field coverage and react to anomalies.

UC-10

Name: Periodic GPS Position Reporting

Actors: System (mobile), Back-Office, Branch Manager

Objectives: Maintain an up-to-date location trail of active agents.

Preconditions: Agent session is open; location permissions granted; tracking is within configured service hours (NFR-10).

Postconditions: Position points are stored and visible on the Back-Office map; historical trail is available.

Nominal Scenario:

1. While the app is in foreground (or allowed background), the system samples GPS/GSM location every 5 minutes.
2. Location + timestamp + agent ID are sent to the server (or queued offline).
3. Back-Office updates the agent's live marker.

Alternative Scenarios:

- *A1 — Location services disabled:* System prompts the agent and may restrict further collections according to policy.
- *A2 — Outside service hours:* Tracking is automatically suspended (NFR-10).

Business Rules: NFR-09 (battery optimization) and NFR-10 apply. BR-Geo-01: Sampling interval is configurable per MFI.

FR-11 — Route Mapping

User Story (US-11): As a Branch Manager, I want to view each agent's historical daily route on a dynamic map so that I can verify that assigned markets were actually covered.

UC-11

Name: Historical Route Visualization Dashboard

Actors: Branch Manager / Administrator (primary), System

Objectives: Reconstruct and display the chronological path followed by an agent on a given day.

Preconditions: Geolocation points have been recorded for the selected agent and date.

Postconditions: Map is rendered with the route polyline, transaction markers and time stamps.

Nominal Scenario:

1. Manager selects agent and date on the Tracking dashboard.
2. System retrieves ordered GPS points and associated transactions.

3. Map displays the route; clicking a point shows time and nearby transactions.

Alternative Scenarios:

- *A1 — Sparse data:* System interpolates or warns that coverage is incomplete.
- *A2 — No data:* Message “No tracking data for this period” is shown.

Business Rules: BR-Route-01: Only managers of the agent’s branch (or higher) can view the route. Data retention follows MFI policy.

FR-12 — Transaction Timestamping & Geotagging

User Story (US-12): As a Risk Officer / Auditor, I want every deposit to be possible only when GPS is active and to be invisibly stamped with precise coordinates and timestamp so that fraudulent or unlocated collections cannot occur.

UC-12

Name: Mandatory GPS Gate & Invisible Transaction Geotagging

Actors: System (automatic), Collection Agent

Objectives: Ensure every collection is authorized only when GPS is active, and attach immutable location and time metadata to every validated deposit.

Preconditions: Agent attempts to start or validate a collection; GPS/location services must be enabled with a valid current fix (BR-05).

Postconditions: Either the collection is blocked with a clear GPS-required message, or the transaction record contains GPS coordinates (lat/lon), accuracy radius and UTC timestamp and is stored for sync.

Nominal Scenario:

1. Before collection entry (or at validation), the system verifies that location services are ON and a valid GPS fix is available.
2. At validation the system captures the current GPS fix and UTC timestamp.
3. Coordinates and timestamp are written into the transaction payload (non-editable by the agent).
4. Data is stored locally (encrypted) and later synced.

Alternative Scenarios:

- *A1 — GPS disabled / permission denied / no fix:* System **blocks** the collection; no amount, denomination or client step proceeds until GPS is activated and a valid fix is obtained.
- *A2 — Fix coarse but valid:* Collection is allowed; accuracy is stored and may be flagged if worse than the configured threshold (e.g. 50 m).

Business Rules: BR-05 (mandatory GPS for collection). BR-GeoTag-01: Geotag is mandatory and non-editable by the agent. BR-GeoTag-02: Accuracy better than 50 m is preferred; coarser but valid fixes are accepted and marked.

FR-13 — Active Geofencing Alert

User Story (US-13): As a Branch Manager, I want an immediate visual alert when an agent conducts a transaction or moves outside the assigned market/district perimeter so that I can intervene quickly.

UC-13

Name: Real-Time Geofence Breach Alert

Actors: System, Branch Manager

Objectives: Detect and notify out-of-zone activity in near real time.

Preconditions: Agent has an assigned geofence (market/district polygon); tracking is active.

Postconditions: Alert appears on the Administrator console; event is logged; optional SMS/push to Manager.

Nominal Scenario:

1. System continuously evaluates agent position against the assigned polygon.
2. On exit (or on transaction outside the zone) an alert event is generated.
3. Back-Office displays a prominent visual alert with agent name, time and map snapshot.

Alternative Scenarios:

- *A1 — Brief excursion:* Configurable grace period (e.g. 2 min) before alert is raised.
- *A2 — Agent returns:* Alert is auto-cleared or marked as resolved.

Business Rules: BR-Fence-01: Geofences are defined per agent or per route by the Administrator. BR-Fence-02: Alerts are retained for audit even after resolution.

FR-14 — Distress / SOS Button

User Story (US-14): As a Collection Agent, I want a discreet emergency button that instantly sends my exact GPS position to the branch administrator so that help can be dispatched in case of assault or threat.

UC-14

Name: Emergency SOS Activation

Actors: Collection Agent (primary), Branch Manager / Administrator, System

Objectives: Transmit an immediate, high-priority location alert without attracting attention.

Preconditions: Agent is authenticated; SOS contacts are configured.

Postconditions: SOS event with GPS is sent via SMS and Internet; Administrator receives visual + audible alert; event is logged as critical.

Nominal Scenario:

1. Agent activates the SOS control (long-press or discrete gesture).
2. System immediately captures current GPS and device status.
3. System sends SMS to predefined numbers and posts an urgent event to the Back-Office.
4. Administrator console flashes a red SOS alert with map and contact options.

Alternative Scenarios:

- *A1 — No network:* SMS is still attempted (cellular); Internet event is queued.
- *A2 — False activation:* Administrator can mark the alert as false positive; agent can cancel within a short window if the UI allows.

Business Rules: BR-SOS-01: SOS cannot be disabled by the agent. BR-SOS-02: Activation is always logged, even if later cancelled.

MODULE 5 — End-of-Day Closing (OFJ) & Reporting

FR-15 — Dynamic Schedule Management

User Story (US-15): As a Branch Manager, I want to configure session opening and closing time windows per branch so that agents can only operate during authorized hours and GPS tracking respects privacy rules.

UC-15

Name: Branch Working-Hours Configuration

Actors: Branch Manager / Administrator

Objectives: Define and enforce the temporal boundaries of collection activity.

Preconditions: Administrator has schedule-management privilege.

Postconditions: Opening/closing times are stored; mobile app enforces the window; GPS tracking is automatically disabled outside the window (NFR-10).

Nominal Scenario:

1. Administrator opens Schedule Configuration for the branch.
2. Administrator sets opening time, closing time and optional grace period.
3. System saves the profile and pushes it to all agents of the branch.
4. Outside the window the mobile app refuses new collections and stops tracking.

Alternative Scenarios:

- *A1 — Agent attempts late collection:* System blocks and displays “Outside authorized hours”.
- *A2 — Emergency override:* Privileged role can temporarily extend the window for a specific agent.

Business Rules: BR-Sched-01: Changes take effect on the next agent login or forced sync.
BR-Sched-02: Historical schedule changes are retained for audit.

FR-16 — End-of-Day Reconciliation

User Story (US-16): As a Branch Cashier, I want a digital reconciliation interface that compares the physical cash I count (by denomination) with the application’s accounting total so that end-of-day closing is fast, accurate and auditable.

UC-16

Name: Digital Cash Desk / OFJ Reconciliation

Actors: Branch Cashier (primary), Collection Agent, System

Objectives: Compute the variance between physical cash and digital records and close the day.

Preconditions: All agents of the branch have synchronized their offline transactions; cashier is authenticated.

Postconditions: Variance Δ is calculated and displayed; session is closed if $\Delta = 0$ or after regularization; CBS export can be triggered.

Nominal Scenario:

1. Cashier selects the agent (or branch aggregate) and opens the OFJ screen.
2. System displays the digital total and expected denomination breakdown.
3. Cashier enters the physical counts by denomination.
4. System computes $\Delta = \text{Physical} - \text{Digital}$ and highlights any difference.
5. If $\Delta = 0$, cashier validates; system closes the session and prepares the accounting export.

Alternative Scenarios:

- *A1 — Negative variance:* System blocks automatic close and opens a variance ticket (links to FR-17).
- *A2 — Positive variance:* System allows close with a surplus note; Manager is notified.

Business Rules: BR-01 (main document) applies. BR-OFJ-01: Closing is impossible while pending offline transactions remain unsynchronized.

FR-17 — Variance Regularization

User Story (US-17): As a Branch Manager / Administrator, I want to record a cash variance as an “agent debt” on the agent’s profile so that shortages are tracked and can be recovered later.

UC-17

Name: Cash Variance Recording as Agent Debt

Actors: Branch Manager / Administrator, System

Objectives: Formalize and persist a shortage against the responsible agent.

Preconditions: An OFJ reconciliation has produced a non-zero variance; Manager has regularization privilege.

Postconditions: Debt amount is posted on the agent’s profile; OFJ can be closed; debt appears in subsequent reports until cleared.

Nominal Scenario:

1. From the OFJ screen the Manager selects “Record as Agent Debt”.
2. System pre-fills the variance amount; Manager adds a comment and confirms.
3. Debt is attached to the agent profile and the OFJ is allowed to close.
4. Agent is notified (optional) of the recorded debt.

Alternative Scenarios:

- *A1 — Partial recovery:* Later, when the agent reimburses, a complementary transaction reduces the outstanding debt.
- *A2 — Write-off:* Privileged role can write off the debt after formal approval workflow.

Business Rules: BR-Var-01: Only negative variances (shortages) can be recorded as agent debt. BR-Var-02: Every debt entry is immutable and fully audited.

FR-18 — Core Banking Export

User Story (US-18): As a Branch Cashier / Accountant, I want the system to generate a daily entries file compatible with major Core Banking systems (Amplitude, FinanSoft, ...) so that collections are automatically posted without manual re-keying.

UC-18

Name: Daily Accounting Export Generation

Actors: Branch Cashier / System Administrator, System, Core Banking System

Objectives: Produce a structured file (or API payload) containing all validated collections of the day ready for CBS import.

Preconditions: OFJ for the day (or branch) has been successfully closed; export format is configured for the target CBS.

Postconditions: Export file is generated and available for download or automatic transmission; export is logged.

Nominal Scenario:

1. After successful OFJ close, Cashier (or automated job) triggers “Generate CBS Export”.
2. System aggregates all validated transactions, applies mapping rules, and produces the file (CSV, XML or proprietary format).
3. File is stored in a secure download area or pushed via API.
4. Cashier confirms receipt / successful import on the CBS side.

Alternative Scenarios:

- *A1 — Mapping error:* System aborts generation and lists the problematic records for correction.
- *A2 — Partial day export:* Administrator can generate an interim export before final close (if policy allows).

Business Rules: BR-Export-01: Only closed and validated sessions are included. BR-Export-02: Export format is selectable per MFI tenant (multi-tenant architecture).

MODULE 6 — Client Digital Activation & Tokenization (Digital Booklet)

FR-19 — Tokenized Digital Activation & Annual Renewal

User Story (US-19a — Field Agent): As a Field Agent, I want to sponsor the activation (or annual renewal) of an existing client’s digital profile by scanning my agent code and triggering the fee split, so that I receive my commission instantly and remain financially incentivized to migrate and retain clients on the mobile platform.

User Story (US-19b — Existing Client): As an existing MFI member, I want to activate my mobile profile using the temporary Activation ID received at account opening and, after paying the digital booklet fee via an agent, obtain a one-year access token, so that I can stop relying on paper booklets while keeping my money fully accessible at the branch.

UC-19

Name: Tokenized Client Activation / Annual Renewal (Digital Booklet)

Actors: Existing Client (primary), Field Agent (sponsor), System, Core Banking System

Objectives: Securely migrate an existing paper-based member into the digital ecosystem, issue a time-limited access token, reward the sponsoring agent, and generate recurring revenue for the MFI while never creating new CBS records.

Preconditions: Client possesses a valid temporary Activation ID issued by the CBS; Field Agent is authenticated on a registered terminal; activation/renewal fee amount is configured.

Postconditions: Client profile is Active with a 365-day access token; fee is split (agent commission + MFI revenue) inside the CBS; temporary Activation ID is invalidated; audit trail is complete.

Nominal Scenario (First Activation):

1. Client installs the app and enters the temporary Activation ID received from the CBS.
2. System queries the CBS; ID is valid. Client is prompted to create permanent credentials (PIN / password / biometric).
3. Profile state is set to *Inactive (Token Required)*. System displays the activation fee screen and requests Field Agent sponsorship.
4. Field Agent scans the client's device QR or enters Agent ID on the client's screen.
5. Client authorizes the fee (cash to agent, Mobile Money, or direct debit from savings with PIN confirmation).
6. System sends split instruction to CBS: commission credited to Agent wallet, remainder to MFI Digital Operations account.
7. System issues a 365-day cryptographic access token, toggles profile to Active, and displays success confirmation. Client can now use full mobile features.

Nominal Scenario (Annual Renewal):

1. On day 366 (or after grace period) the token expires. App enters read-only mode: balances remain visible; transactional operations are locked. Message: "Votre carnet numérique annuel est arrivé à expiration. Votre argent reste en sécurité. Veuillez contacter votre agent de collecte de confiance pour renouveler votre accès de proximité."
2. Client meets Field Agent; Agent sponsors the renewal fee (same split mechanism).
3. System issues a new 365-day token and restores full transactional access.

Alternative Scenarios:

- *A1 — Activation ID not found / invalid / expired:* System displays "ID introuvable. Vous n'êtes pas encore inscrit dans nos registres. Veuillez vous rapprocher d'une agence ou d'un agent de collecte pour ouvrir votre compte." No profile is created. User is returned to landing screen.
- *A2 — Profile already Active:* System bypasses fee and redirects to normal login.
- *A3 — Grace period (days 366–380):* Warning banner is shown but full features remain available; Agent is notified during market visits.
- *A4 — Payment failure:* Activation remains blocked; fee is not split; client and agent are informed; retry is possible.
- *A5 — Agent not authenticated / invalid Agent ID:* Sponsorship is rejected; activation cannot proceed.

Business Rules: BR-04 applies in full. BR-Token-01: Money is never locked — only mobile transactional inputs are restricted. BR-Token-02: Renewal fee may be deducted from the client's own savings after PIN confirmation (zero out-of-pocket option). BR-Token-03: Temporary Activation ID is single-use and destroyed after credential creation. BR-Token-04: Token validity and fee amounts are configurable per MFI tenant.

FR-20 — Client Profile Consultation

User Story (US-20): As an activated Client, I want to consult my personal profile (name, membership number, phone, branch, token status and expiry) so that I always know the state of my digital booklet and can prove my membership when needed.

UC-20

Name: Client Profile Self-Service View

Actors: Client (primary), System

Objectives: Provide the client with a clear, always-available view of his/her identity and digital-booklet status.

Preconditions: Client is authenticated (or in read-only mode after token expiry); profile has been activated at least once.

Postconditions: Profile data is displayed; no modification is performed (consultation only).

Nominal Scenario:

1. Client opens the mobile application and authenticates (or lands in read-only mode).
2. Client navigates to the “My Profile” / “Mon Profil” section.
3. System retrieves and displays: full name, membership ID, registered phone, assigned branch, activation date, current token status and expiry date.
4. Client can return to the dashboard or other self-service screens.

Alternative Scenarios:

- *A1 — Offline:* System displays last synchronized profile snapshot and clearly indicates that data may not be up to date.
- *A2 — Token expired:* Profile remains fully visible; a prominent banner indicates that renewal is required for transactional features.

Business Rules: BR-Client-01: Profile data is read-only for the client; only the MFI Administrator can modify identity attributes. BR-Token-01 still applies (money never locked).

FR-21 — Account Balance & Summary

User Story (US-21): As an activated Client, I want to see my current savings balance and year-to-date cumulative contributions so that I have continuous confidence that my money is safe and correctly recorded.

UC-21

Name: Client Account Balance & Summary Consultation

Actors: Client (primary), System, Core Banking System (for live balance)

Objectives: Give the client real-time (or near-real-time) visibility of his/her financial position.

Preconditions: Client is authenticated or in read-only mode; at least one account is linked to the digital profile.

Postconditions: Balance and summary figures are displayed; no debit or credit is performed.

Nominal Scenario:

1. Client opens the “My Account” / “Mon Compte” section.
2. System queries the platform cache and, when online, the Core Banking System for the

current balance.

3. System displays: current balance, cumulative deposits YTD, number of contributions, last contribution date and amount.
4. Client can pull-to-refresh to force a live query when connectivity is available.

Alternative Scenarios:

- *A1 — Offline:* Last known balance is shown with a clear “Dernière synchronisation” timestamp.
- *A2 — Token expired:* Balance remains visible (read-only). Transactional buttons are disabled.
- *A3 — CBS unreachable:* System falls back to cached balance and notifies the client.

Business Rules: BR-Client-02: Balance visibility is never blocked by token expiry. BR-Client-03: Only positive, non-sensitive summary data is shown; full ledger remains on the MFI side.

FR-22 — Contribution History & Mini-Statement

User Story (US-22): As an activated Client, I want to consult the full history of my contributions (and receive a short mini-statement by USSD or SMS) so that I can replace the paper booklet pages with a reliable digital record.

UC-22

Name: Contribution History & Mini-Statement

Actors: Client (primary), System

Objectives: Reproduce and improve the proof-of-deposit function of the physical booklet.

Preconditions: Client is authenticated or in read-only mode; contribution records exist.

Postconditions: History list or mini-statement is displayed / sent; no data is modified.

Nominal Scenario (Mobile App):

1. Client opens the “History” / “Historique” section.
2. System presents a chronological list of contributions (date, amount, agent code, optional location).
3. Client can filter by period (last 7 / 30 / 90 days or custom range) and export or share a simple PDF/image summary if desired.

Nominal Scenario (USSD / Feature Phone):

1. Client dials the short code (e.g. *XXX#).
2. System authenticates via registered phone number + optional PIN.
3. System returns the last 5 contributions as a concise USSD menu or Flash SMS.

Alternative Scenarios:

- *A1 — No history yet:* Friendly empty-state message is shown.
- *A2 — Offline (app):* Cached history is displayed with sync timestamp.
- *A3 — Token expired:* Full history remains consultable; only new deposits are blocked.

Business Rules: BR-Client-04: History is append-only from the client perspective. BR-Client-05: USSD/SMS mini-statement is limited to the last N transactions (configurable, default 5) to respect message size and cost.

The user stories and use cases above are fully aligned with the functional requirements FR-01 to FR-22 and with the business rules already defined (BR-01 to BR-05) and the non-functional constraints (NFR-01 to NFR-12). They constitute the detailed behavioral specification that will guide both development and acceptance testing of the MICROFI MVP and subsequent releases. The Client Module (FR-19 to FR-22) elevates MICROFI from a pure agent collection tool to a complete digital booklet ecosystem that preserves agent incentives while giving members continuous visibility and proof of their savings.

Detailed Non-Functional Requirements (NFR)

SECURITY & REGULATORY COMPLIANCE

NFR-01

Data Encryption — All data stored locally on the agent's smartphone in Offline mode shall be encrypted to the AES-256 standard.

NFR-02

COBAC & BEAC Compliance — The architecture shall comply with anti-money laundering (AML) and counter-terrorism financing (CFT) directives.

NFR-03

Personal Data Protection — The system shall store data in accordance with data protection legislation (secure hosting, restricted access).

PERFORMANCE & USABILITY

NFR-04

Field Entry Time — The collection interface shall be optimized so that a complete transaction (identification + denomination breakdown) takes less than 30 seconds.

NFR-05

SMS Latency — The SMS gateway shall deliver the client notification within a maximum of 15 seconds after validation (in a connected area).

NFR-06

Inclusive Visual Interface — The mobile application shall use clear color coding and explicit icons to remain usable by users with low literacy levels.

ROBUSTNESS & AVAILABILITY

NFR-07

Server Availability — The central Back-Office hosted in the Cloud shall maintain an availability rate of 99.9%.

NFR-08

Post-Outage Synchronization — Upon restoration of internet connectivity, the mobile application shall synchronize pending transactions transparently in the background, without blocking the agent.

NFR-09

Battery & GPS Optimization — The tracking module shall use a hybrid algorithm (GPS / GSM triangulation) to limit energy consumption and guarantee at least 8 continuous hours of field autonomy.

NFR-10

Privacy / Off-Duty Compliance — The agent's GPS tracking shall technically and automatically deactivate outside the service hours configured by the MFI administrator.

SCALABILITY & INTEROPERABILITY

NFR-11

Multi-Tenant Architecture — The platform shall be capable of hosting multiple MFIs in complete, fully partitioned isolation from one another.

NFR-12

API Agnosticism — The system shall provide an open API layer to easily interface with any payment aggregator or Core Banking system on the market.

Operational RACI Matrix of the System

This matrix defines governance and accountability within each MFI for the execution of the critical modules configured in the application.

Legend: R = Responsible for the action, A = Approves (final accountable party), C = Consulted, I = Informed.

Feature Priority Matrix & MVP Derivation

With a hard production window of **six (6) weeks**, MICROFI cannot launch with all 18 functional requirements at once. Rather than cutting scope subjectively, each FR is scored against four weighted criteria to produce an objective, defensible MVP boundary.

Operational Activity	Branch Manager (Admin)	Collection Agent	Risk Director / CEO	Technical Support (PK Fokam Ctr.)
Agent profile creation & validation	A/R	I	C	C
Dynamic schedule & OFJ configuration	R	I	A	C
Collection ceiling modification	R	I	A	C
Tracking alert handling (SOS / Out-of-Zone)	A/R	I	C	I
Physical cash handover validation	A/R	R	I	I
Corrective maintenance & API evolution	I	I	I	A/R

Table 4: Operational RACI Matrix — System Modules

Scoring Criteria & Weights

- **Criticality(C) (35%)** — Business/risk impact if the feature is missing. 1 = nice-to-have; 5 = blocking or core to the value proposition.
- **Regulatory & Compliance Urgency(R) (15%)** — Direct link to COBAC/BEAC obligations (traceability, LBA/FT, daily reconciliation). 1 = no regulatory link; 5 = directly mandated.
- **Development Speed(S) (30%)** — Inverse of estimated effort. Given the 6-week ceiling, speed is weighted almost as heavily as criticality. 1 = >5 dev-days / multi-system integration; 5 = ≤2 dev-days, self-contained.
- **Cost Efficiency() (20%)** — Inverse of financial expenditure (external fees, licensing, gateways). 1 = high recurring external cost; 5 = negligible cost.

Weighted Score = $0.35 \times \text{Criticality} + 0.15 \times \text{Compliance} + 0.30 \times \text{Dev Speed} + 0.20 \times \text{Cost Efficiency}$, each criterion rated 1–5.

Decision Thresholds

- **Score ≥ 3.9** — **MVP (V1.0)**, built within the 6-week window.
- **Score 3.0 – 3.89** — **Fast-Follow (V1.1)**, immediately after MVP launch.
- **Score < 3.0** — **Backlog (V2)**, revisited in a later release.

Scored Requirements Matrix

Effort is estimated in person-days for a single full-stack developer; C = Criticality, R = Regulatory Urgency, S = Dev Speed, \$ = Cost Efficiency (all rated 1–5).

ID	Feature	Effort (days)	C	R	S	\$	Score	Decision
FR-01	Single Sign-On Authentication	3	5	4	4	5	4.55	MVP
FR-02	Admin Enrollment (Agent CRUD)	2	5	3	5	5	4.70	MVP

ID	Feature	Effort (days)	C	R	S	\$	Score	Decision
FR-03	Multi-Bank Top-Up	6	3	2	1	2	2.05	Backlog*
FR-04	Dynamic Escrow Lockout	2	5	4	5	5	4.85	MVP
FR-05	Exceptional Ceiling Adjustment	2	3	2	5	5	3.85	Fast-Follow
FR-06	Multi-Method Client Identification (QR, ID, phone, registry search)	3	5	4	4	5	4.55	MVP
FR-07	Offline Collection Mode	5	5	3	2	4	3.60 [†]	MVP [†]
FR-08	Full Denomination Break-down	3	5	5	4	5	4.70	MVP
FR-09	Multi-Channel Notification	4	4	3	3	3	3.35	Fast-Follow
FR-10	Real-Time Geolocation	1	4	3	5	4	4.15 [‡]	MVP
FR-11	Route Mapping Dashboard	4	3	2	3	3	2.85	Backlog
FR-12	Mandatory GPS + Geotagging	1	5	5	5	5	5.00	MVP
FR-13	Active Geofencing Alert	4	3	3	3	3	3.00	Fast-Follow
FR-14	Distress / SOS Button	2	5	2	5	4	4.35	MVP
FR-15	Dynamic Schedule Management	3	4	2	4	5	3.90	MVP
FR-16	End-of-Day Reconciliation	4	5	5	3	4	4.20	MVP
FR-17	Variance Regularization	2	4	3	5	5	4.35	MVP
FR-18	Core Banking Export	3	5	4	4	4	4.35	MVP
FR-19	Tokenized Digital Activation (Digital Booklet)	6	5	3	2	3	3.40	Fast-Follow
FR-20	Client Profile Consultation	2	4	2	5	5	4.05	MVP
FR-21	Account Balance & Summary	3	5	2	4	4	4.05	MVP
FR-22	Contribution History & Mini-Statement	3	4	2	4	4	3.70	Fast-Follow

Table 5: Feature Priority Matrix — All 22 Functional Requirements Scored

*FR-03 backlog rating applies to the **full** multi-bank/interbank (GIMAC) automation. A minimal single-channel, cashier-recorded top-up ships bundled inside FR-04 at negligible extra cost, so escrow remains usable in V1.0.

[†]FR-07 is promoted to MVP by architecture-dependency override, not by raw score alone — see note below.

[‡]FR-10's 1-day estimate covers basic periodic capture-and-send via native platform APIs (FusedLocationProviderClient / CLLocationManager) while the app is active. Hardening it to survive 8 hours of background execution across OEM battery restrictions is scoped under NFR-09 and refined post-launch.

ANALYST OVERRIDE NOTE

FR-07 (Offline Collection Mode) scores below the 3.9 MVP threshold once Development Speed is factored in, because building reliable local encrypted storage and background sync takes real engineering time. However, offline capability is the *architectural foundation* the entire platform is built on — collection genuinely happens in low-connectivity markets, and FR-06 (all client identification channels), FR-08, and FR-12 all assume it. A purely mechanical score cannot be allowed to cut a load-bearing dependency; FR-07 is therefore force-included in MVP regardless of its numeric ranking. This is the one deliberate exception to the scoring model in this release.

Resulting Scope: MVP (V1.0) vs Fast-Follow (V1.1) vs Backlog (V2)**✓ MVP SCOPE — V1.0 (WEEKS 1–6)**

FR-01, FR-02, FR-04, FR-06, FR-07[†], FR-08, FR-10[‡], FR-12, FR-14, FR-15, FR-16, FR-17, FR-18, FR-20, FR-21

Single sign-on, agent enrollment, escrow-based lockout, multi-method client identification, offline mode, full denomination breakdown, basic real-time GPS, transaction geotagging, SOS button, schedule management, end-of-day reconciliation, variance regularization, Core Banking export, plus Client Profile consultation and Account Balance view — the complete Digital Cash Desk loop with essential client self-service visibility.

🔧 FAST-FOLLOW — V1.1 (IMMEDIATELY POST-LAUNCH)

FR-05, FR-09, FR-13, FR-19, FR-22

Remote ceiling override, SMS/Bluetooth notification polish, active geofencing alerts, Tokenized Digital Activation / Digital Booklet (agent-sponsored activation + annual renewal + commission split), and Contribution History / USSD mini-statement. Also includes the NFR-09 hardening pass for 8-hour background GPS tracking. These features complete the client digital booklet experience and strengthen agent incentives.

✗ BACKLOG — V2

FR-03 (full multi-bank/interbank top-up automation), FR-11 (route-mapping dashboard)

Consistent with the earlier out-of-scope items already identified (real-time Mobile Money gateways, predictive ML fraud detection, dedicated client app, automated multi-switch GIMAC integration).

Capacity check: The MVP scope represents approximately **33 person-days** of functional development, plus an estimated 8–10 person-days for baseline security and compliance implementation (NFR-01–03, treated as non-negotiable regardless of scoring), integration, and quality assurance.

The project is executed by a **single full-stack developer** supported by AI-assisted development tools, automated code generation, and productivity-enhancement workflows.

Over a 6-week development window, this approach provides an estimated effective capacity of around **60 person-days**, allowing coverage of the approximately **41–43 person-day MVP effort** while maintaining a reasonable buffer for testing, debugging, integration challenges, and technical adjustments.

Alignment with the Institution's S.M.A.R.T. Objectives

The adoption of the MICROFI platform allows partner MFIs to achieve precise, measurable objectives:

- **Specific** — Fully digitize the proximity collection cycle and branch-level reconciliation.
- **Measurable** — Reduce counterfeit-note disputes by 90% and bring the end-of-day reconciliation time down to under 15 minutes.
- **Achievable** — Through a hybrid (offline-first) architecture adapted to the realities of both Category 1 and Category 2 networks.
- **Realistic** — Designed specifically for the regulatory and operational context of the CEMAC zone.
- **Time-bound** — MVP built within a 6-week production window and deployed in a pilot phase before broader rollout.

5. Conclusion

The analysis conducted in this document has highlighted the major limitations of the current proximity collection process used by microfinance institutions: strong dependence on manual operations, data entry errors, difficulties in end-of-day reconciliation, lack of transaction traceability, and limited operational visibility over field agents' activities.

In response to these challenges, the proposed MICROFI solution provides an approach adapted to the operational realities of microfinance institutions in the CEMAC region. By relying on a *digital-first* and *offline-first* approach, the platform ensures business continuity even in environments characterized by limited connectivity. The transformation from the AS-IS process to the TO-BE process therefore introduces a higher level of control, security, and efficiency.

The integration of features such as secure agent enrollment, guarantee and collateral management, transaction geolocation, real-time collection monitoring, automated reconciliation, and export capabilities toward existing banking systems addresses the main business requirements identified. The solution also improves governance through a clear definition of responsibilities among stakeholders according to a RACI approach.

The decision to adopt an incremental development strategy through an MVP makes it possible to control project risks while delivering operational value quickly to partner institutions. This first version provides an evolutionary foundation capable of supporting future advanced features such as intelligent fraud detection using machine learning, complete integration with Mobile Money networks, and predictive analysis of agent performance.

Therefore, MICROFI is not merely a collection digitization tool, but a true operational transformation platform enabling microfinance institutions to improve efficiency, strengthen customer trust, and achieve better risk management.

5. Glossary

Alphabetical list of abbreviations and key terms used in the Business Analysis and in the System Architecture and Software Design Document.

ACID *Atomicity, Consistency, Isolation, Durability*. Set of properties that guarantee reliable database transactions (used for financial ledger modules via Spring Data JPA).

AES *Advanced Encryption Standard*. Symmetric encryption algorithm; MICROFI uses AES-256 for offline local storage on agent terminals (NFR-01).

AML *Anti-Money Laundering*. Legal and operational measures to prevent financial institutions from being used for money laundering.

AMQP *Advanced Message Queuing Protocol*. Messaging protocol used by RabbitMQ to decouple the Core Backend from the CBS Middleware.

API *Application Programming Interface*. Rules and protocols allowing software systems to communicate and exchange data.

AS-IS Representation of the current state of a business process before transformation (existing practices, workflows and limitations).

BEAC *Bank of Central African States*. Central bank for monetary policy and financial regulation in the CEMAC region.

BOLA *Broken Object-Level Authorization*. Access-control flaw where a caller can access or modify objects belonging to another user or branch; addressed by server-side branch scoping.

BR *Business Rule*. Constraint or policy that the system must enforce (e.g. BR-01 OFJ delta, BR-03 escrow lock, BR-04 digital booklet token, BR-05 mandatory GPS).

Branch Cashier Staff member responsible for physical cash at the branch, counting, reconciliation and validation of daily collections.

CBS *Core Banking System*. Central system managing customer accounts, ledgers and financial operations (e.g. Amplitude, FinanSoft).

CEMAC *Economic and Monetary Community of Central Africa*. Regional community with shared monetary and financial regulations.

CFT *Counter-Terrorism Financing*. Measures to prevent financing of terrorism through the financial system.

COBAC *Central African Banking Commission*. Supervisory authority for banking regulation and prudential control in CEMAC.

CRUD *Create, Read, Update, Delete*. The four basic operations on data entities.

DDL *Data Definition Language*. SQL statements that define schema objects (tables, indexes, constraints).

Digital Cash Desk Digital process that replaces manual cash reconciliation by comparing recorded collections with physical cash (OFJ).

Digital Escrow Electronic financial guarantee that limits the volume of cash an agent may carry before a top-up is required.

DTO *Data Transfer Object*. Object used to carry data across process or service boundaries without exposing internal entities.

FK *Foreign Key*. Database constraint linking a column to the primary key of another table.

FR *Functional Requirement*. Requirement describing a function or service the system must provide (FR-01 to FR-22).

Geofencing Location-based control that defines virtual geographic boundaries and raises alerts when a device enters or leaves them.

Geolocation Identification of the geographic position of a device or transaction (GPS, network-based, etc.).

GIMAC *Groupement Interbancaire Monétique de l'Afrique Centrale*. Regional interbank payment network in the CEMAC zone.

GPS *Global Positioning System*. Satellite positioning used for agent tracking and transaction geo-tagging.

Guarantee / Collateral Financial or contractual mechanism covering potential losses from funds entrusted to a collection agent.

HMI *Human–Machine Interface*. Design of interactive interfaces (agent app, client app, back-office).

IMEI *International Mobile Equipment Identity*. Unique device identifier used to bind an agent session to an enrolled terminal (FR-01).

JPA *Jakarta Persistence API* (formerly Java Persistence API). Standard for object–relational mapping; used with Spring Data JPA for ACID financial modules.

JSON / JSONB *JavaScript Object Notation / binary JSON* in PostgreSQL. Structured payload and storage format (e.g. denomination details, outbox jobs).

JWT *JSON Web Token*. Signed token used for stateless authentication of agents, clients and back-office users after login.

LBA/FT *Lutte contre le Blanchiment d'Argent et le Financement du Terrorisme*. French CEMAC terminology equivalent to AML/CFT.

LTS *Long-Term Support*. Designation for a stable software release line with extended maintenance (e.g. Java 21 LTS).

MC² *Mutuelle Communautaire de Croissance*. Community-based microfinance model in Cameroon.

MFI / IMF *Microfinance Institution*. Institution providing savings, credit and related services to underserved populations.

MICROFI Name of the mobile microfinance collection and digital booklet platform specified in this documentation.

MVP *Minimum Viable Product*. First operational release containing only the features required for pilot deployment and validation.

- mTLS** *Mutual Transport Layer Security*. TLS mode in which both client and server authenticate with certificates.
- NFR** *Non-Functional Requirement*. Requirement on quality attributes (security, performance, availability, etc.).
- OFJ** *Opération de Fin de Journée*. End-of-day closing: reconciliation of digital totals with physical cash and recording of variances.
- Offline-first** Design approach where the edge application works without continuous connectivity and synchronises when the network returns.
- OTLP** *OpenTelemetry Protocol*. Standard protocol used to export traces (and metrics/logs) to backends such as Tempo.
- PII** *Personally Identifiable Information*. Data that can identify an individual (name, phone, membership number, etc.).
- PIN** *Personal Identification Number*. Secret numeric code used by agents or clients to authorize sensitive actions.
- POS** *Point of Sale*. Physical payment terminal for transactions and receipt printing.
- QR Code** *Quick Response Code*. Two-dimensional barcode scanned to identify clients or resources.
- RACI** *Responsible, Accountable, Consulted, Informed*. Framework clarifying stakeholder roles on each activity.
- RBAC** *Role-Based Access Control*. Authorization model granting permissions according to assigned roles (admin, cashier, manager).
- Reconciliation** Comparison of system-recorded transactions with physical cash to detect and resolve discrepancies (Δ).
- REST** *Representational State Transfer*. Architectural style for HTTP APIs used by the Core Backend and Middleware surfaces.
- SAD** *System Architecture and Software Design Document*. Technical design document that realises the Business Analysis requirements.
- SMS** *Short Message Service*. Short text notifications to mobile users (e.g. deposit confirmation).
- SOS** Emergency alert function that sends the agent's location and distress signal to predefined contacts and the back-office (FR-14).
- SSO** *Single Sign-On*. Authentication mechanism providing access after one credential verification (agent IMEI-bound SSO).
- STRIDE** Threat-modelling framework: *Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege*.
- TLS** *Transport Layer Security*. Cryptographic protocol securing data in transit (API Gateway, service links, brokers).
- TO-BE** Target state of a process after improvement (expected future operating model).
- TPU** *Terminal Portable Utilisateur*. Secure portable terminal assigned to a field collection agent.

Traceability Ability to retain a complete history of transactions, actions and events for audit and monitoring.

TTL *Time To Live*. Validity duration of a token, cache entry or session before expiry.

UC *Use Case*. Structured description of an interaction between actors and the system to achieve a goal (UC-01 to UC-22).

US *User Story*. Short requirement form “As a [role], I want [goal] so that [benefit]”.

USSD *Unstructured Supplementary Service Data*. GSM protocol for menu-driven services via short codes without Internet data.

UUID *Universally Unique Identifier*. 128-bit identifier used as primary key type in the MICROFI schema.

Wallet Digital account or store holding funds, guarantees or transactional balances (e.g. agent escrow wallet).

XAF *Central African CFA franc*. Currency used for all monetary amounts in MICROFI (stored as integer minor units).